Online Food Delivery Analysis: Data-Driven Business Insights

Customer & Order Analysis.

7.Average delivery time by city

Delivery time is essentially the same in every city. Fastest city is Bangalore and Chennai and slowest is Mumbai. The gap is 2% of the average.

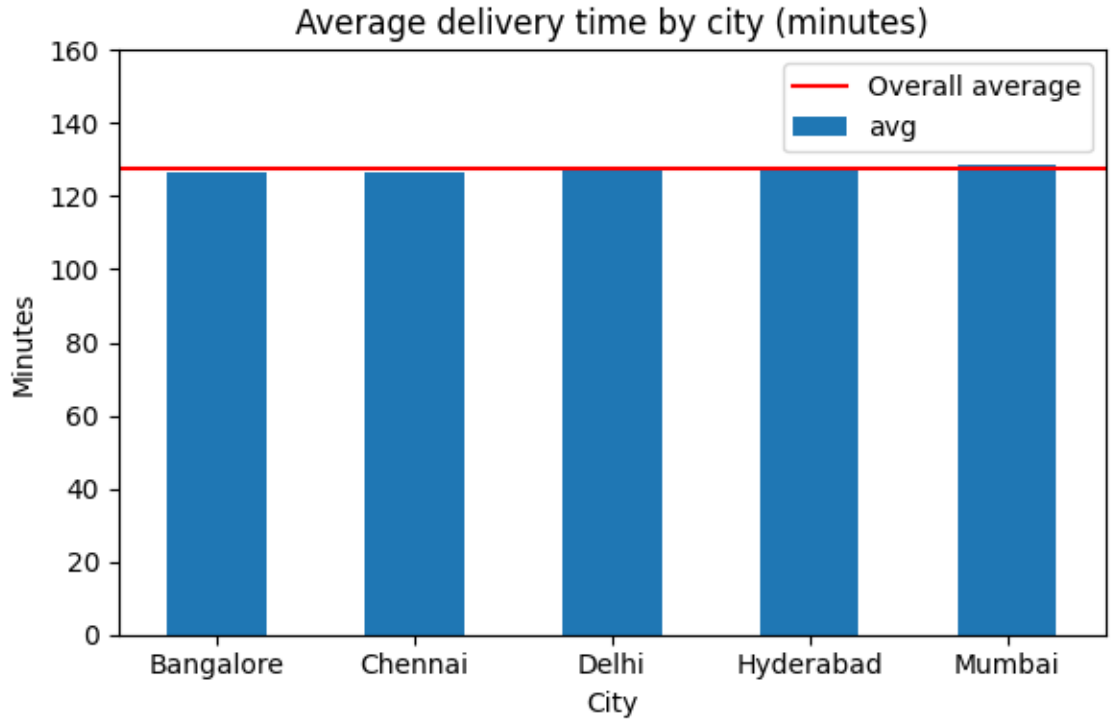
Illustrative sample records :

City	orders	avg	median
Bangalore	9526	126.5	70.0
Chennai	9392	126.5	70.0
Delhi	9403	127.6	121.0
Hyderabad	9557	127.8	120.0
Mumbai	9393	128.6	122.0

2 . VISUALIZATIONS

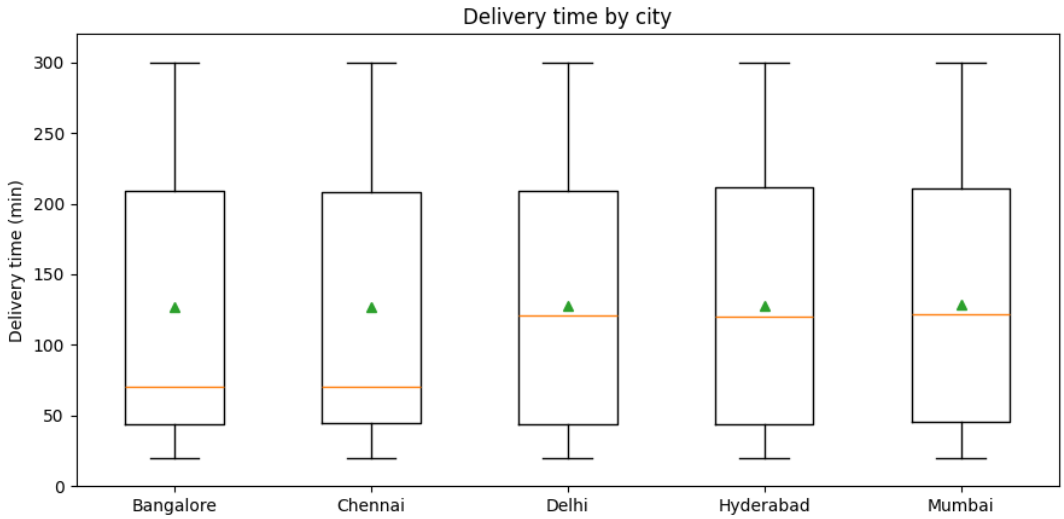
Visualization – 1

Bar chart shows average delivery time by city.



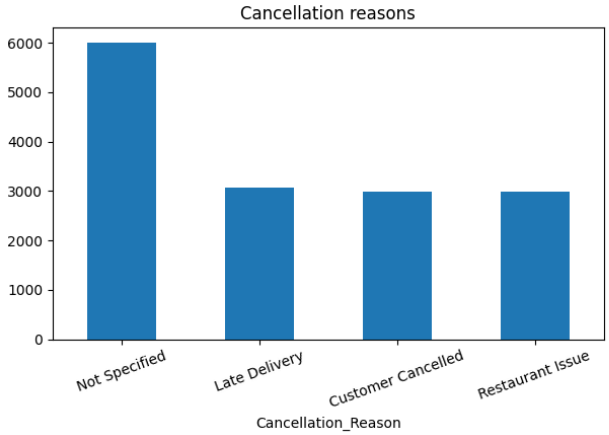
Visualization – 2

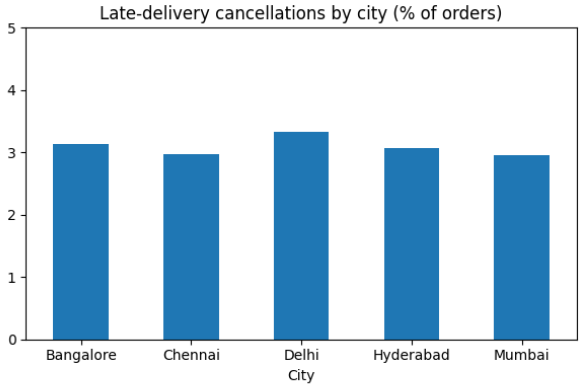
Below boxplot shows delivery time by city .



Visualization – 3

This bar chart shows late delivery cancellation reason.





3.ANALYTICAL INSIGHTS – DEMAND AND REVENUE

INSIGHT – 1: Average delivery time is almost the same in every city.

It ranges from 126 to 128 minutes, around an overall 127. The 2.2-minute gap between cities is less than 2% and not statistically significant , since each city's average is itself uncertain by about ±2 minutes.

INSIGHT – 2: The split between fast and slow deliveries is the same everywhere.

About 40% of deliveries in each city take 60 minutes or less, and 41% to 42% take more than 150 minutes . Every city has the same mix of quick and very slow deliveries.

INSIGHT – 3: Cancellations don't follow speed.

The cancellation rate is 14.9% to 15.2% in every city , and Mumbai is the slowest. Late-delivery cancellations are 2.95% to 3.32% of orders, with Delhi highest even though its speed is in the middle.

INSIGHT – 4: The "fastest city" changes every month.

Chennai was fastest in 5 months, Bangalore and Hyderabad in 3 each and Delhi in 1. Each city's monthly average swings by 9 to 13 minutes, which is bigger than the gap between cities, so no city is consistently faster.

4.ANALYTICAL INSIGHTS-OPERATIONS AND CUSTOMER EXPERIENCE:

INSIGHT – 5: Average Delivery Time by City

Identify cities with the longest delivery times and prioritize operational improvements.

INSIGHT – 6: Delivery Time vs. Order Volume

Optimize staffing and order allocation in high-demand cities. This can help to decrease delay in delivery.

INSIGHT- 7: Improve monthly Delivery Time Trends

Monitor operational changes and establish continuous improvement targets. This operation to done by every cities to improve and analyse the delivery time .